

RECENT ADVANCES IN X-RAY MICROSCOPY INSTRUMENTATION DEVELOPMENTS AT NSLS-II*

E. Nazaretski[†], Weihe Xu, D. S. Coburn, H. Yan, Z. Gao, X. Huang, Wei Xu, N. Bouet, J Zhou, M. Ge, W.-K. Lee, Y. Yang, A. M. Kiss, Y. S. Chu

National Synchrotron Light Source II, Brookhaven National Laboratory, Upton, NY, USA

Abstract

X-ray microscopy is a mature characterization tool routinely used to investigate diverse material questions of science, technology, and engineering. The high penetration power of X-rays allows the utilization of different characterization methods and reveals elemental composition, crystalline phases, strain distribution, oxidation states, etc., in macroscopic and microscopic samples. Full-field and scanning X-ray microscopes serve similar scientific purposes but provide technical capabilities that complement each other. In recent years, several X-ray microscopy systems have been designed, constructed, and commissioned at NSLS-II. Below, we provide a brief technical overview of recently designed microscopy instruments. It includes some design details of the Multilayer Laue Lens (MLL) - based nanoprobe optimized for ~ 10 nm spatial resolution imaging; the zone plate (ZP)-based full-field imaging system capable of 1-minute nano-tomography measurements; and a new Kirkpatrick-Baez (KB)-based scanning microscope designed for ~ 200 nm spatial resolution experiments.

INTRODUCTION

NSLS-II is a 3 GeV synchrotron facility with a strong emphasis on imaging and microscopy applications. There is a dedicated beamline imaging and microscopy program (IMP) that comprises a few scanning and full-field imaging technique beamlines. The majority of the high-resolution endstation microscopes have been designed and constructed in-house, with technical details for some of them elucidated in the sections below.

MLL-BASED HARD X-RAY SCANNING MICROSCOPE

The 3-ID Hard X-ray Nanoprobe (HXN) beamline enables fluorescence, tomography, differential phase contrast, diffraction, nano-spectroscopy, and ptychography as imaging modalities in 2D/3D, and the photon energy range of 6 to 18 keV. It routinely delivers a focus size of ~ 10 nm using MLL optics and ~ 40 nm using ZPs [1-4]. The endstation microscope, shown in Fig. 1, has been designed and constructed in-house and became available to general users around 2015 [5, 6]. A few prototype systems have been commissioned and evaluated for performance prior to finalizing the design of the HXN endstation [7-9]. MLLs were proposed as an alternative to the ZP optics,

overcoming the aspect ratio problem, in the hard X-ray regime [10-12]. A pair of linear MLLs is needed to achieve point focusing; therefore, orthogonality between the two MLLs is crucial since any angular misalignment introduces an additional cross-phase term in the wavefront, causing blurring of the focus. Also, placement of individual MLLs along the X-ray beam is important (within the optics depth of focus) since inaccuracies introduce astigmatism. For a ~ 10 nm point focus, the angular misalignment should be better than 0.01° , and placement along the beam direction should be within $\pm 2 \mu\text{m}$ [13]. Also, a typical focal length for a $50 \mu\text{m}$ aperture, ~ 10 nm resolution lens is less than 5 mm, making it challenging to design such a microscope while ensuring both MLL and ZP compatibility. Therefore, as shown in the lower panel of Fig. 1, two individual modules, i.e., the MLL module and the ZP module, were designed, constructed, and installed inside the vacuum chamber; they can be hot-swapped during the beamtime cycle without any major interruptions to user operation.

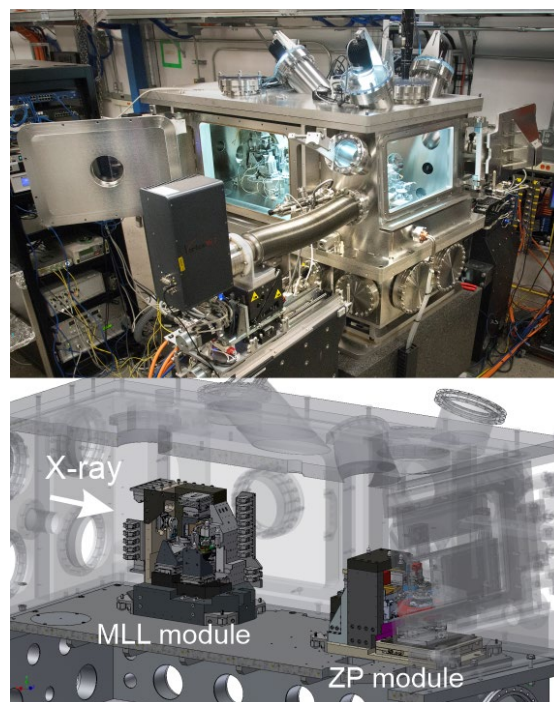


Figure 1: Top panel: a photo of the MLL/ZP microscope installed at 3-ID. Lower panel: upper-level Computer-Aided Design (CAD) model of the MLL and ZP modules arrangement inside the vacuum chamber.

In total, the MLL module comprises 21 rotational and translational degrees of motion and incorporates 9 fiber-optic interferometer channels, which were particularly

* This Work is Not Export Controlled

[†] enazaretski@bnl.gov

useful during the commissioning phase of the instrument. Due to the form factor limitations, many Attocube, SmarAct, PI, and Piezosystem Jena piezo-driven stages were used. The choice of piezo-driven stages was mainly due to their compact size and minimal heat dissipation when holding the target position. The fundamental resonance frequencies for the fully loaded sample stages are ~ 170 Hz, while the vertical and horizontal MLL optics assemblies are in the range of 320 and 350 Hz. The system can be operated in a high-vacuum environment ($\sim 10^{-6}$ mbar); however, during routine user measurements, the vacuum chamber is typically backfilled with ~ 325 mbar of He gas for faster thermalization of the samples. Recently, the next-generation MLL/ZP-suitable microscope has been designed and commissioned at HXN. It is more universal than its predecessor and can accommodate both 1D and 2D diffraction optics. It uses more precise line-focusing interferometry for the actual sample position encoding and is capable of acquiring sub-10 nm data with $\sim$ kHz range speed. The newly developed system will serve as a foundation for the HXN end station upgrade that is envisioned in the near future [14].

KB-BASED HARD X-RAY SCANNING MICROSCOPE

The 5-ID Submicron Resolution X-ray Spectroscopy (SRX) beamline fills the resolution gap between the microprobe beamlines and the nanoscale imaging beamlines at NSLS-II; it delivers to users ~ 200 nm focus in the energy range of 4.5 – 20 keV. The home-built KB-based microscope, shown in Fig. 2, consists of the in-vacuum KB optics module and the sample stage outside the vacuum chamber [15]. Both 2D and 3D scanning fluorescence and spectroscopy capabilities are available [16, 17]. The KB mirror substrates were fabricated by JTEC (the metal coatings were deposited at the NSLS-II R&D Lab), have a length of 200 mm (vertical) and 100 mm (horizontal), and the measured slope errors of ~ 70 nrad and ~ 50 nrad, respectively. The KB-optics mechanism is mounted on the Invar-36 base plate attached to the granite through the bellows coupling. Two independent motion stacks are used for alignment of the vertical and horizontal focusing mirrors. The customized flexures fabricated by Electrical Discharge Machining (EDM) are used for fine angular adjustments. After several iterations and experimental tests, the final flexure assembly has a resonance frequency of ~ 250 Hz, making it less sensitive to the cultural noise. While standard in-vacuum stepper motors are used for coarse positioning, a pair of PI Nexline and piezo-expansion actuators is used for fine pitch control of each X-ray mirror. In addition to the native optoelectronic encoders of the actuators, an external Invar-based reference frame inside the vacuum chamber accommodates a set of fiber-optic interferometer heads, which are used to monitor the actual position of the mirrors. The sub-50 nrad pitch motions were demonstrated during the commissioning phase of the instrument. The X-ray beam exits the vacuum chamber through a 50 μ m-thick diamond window. During the design phase of the sample

stage, a strong emphasis was given to fast scanning and the ability to perform tomographic measurements. Due to geometrical limitations of the available space envelope, the use of high-quality large-form-factor air-bearing stages was not possible; therefore, a much smaller piezo-driven SmarAct rotary stage was used in combination with a diamond-turned reference ring and interferometric line-focusing position reading, similar to the approach described in [18]. Scanning is performed using a standard nPoint 100 μ m travel range 3D scanner with a customized XYZ coarse stage underneath that is based on the SmarAct long travel range CLL-series actuators. During the initial commissioning of the instrument, the resolution of ~ 188 nm in the horizontal direction was demonstrated with a corresponding photon flux in the focal spot of $\sim 3.3 \times 10^{10}$ photon/s.

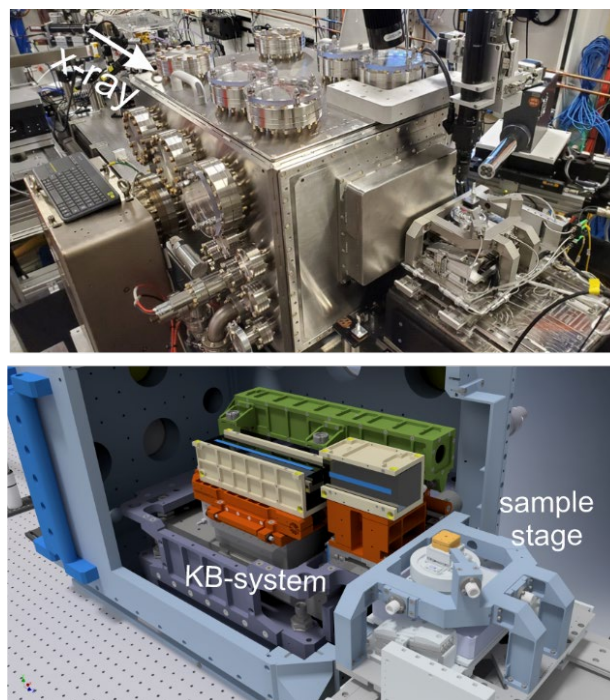


Figure 2: Top panel: a photo of the KB microscope installed at 5-ID. Lower panel: upper-level CAD model of the KB mechanism and sample stage arrangement.

FULL-FIELD ZP-BASED MICROSCOPE

The 18-ID beamline enables tomographic full-field and X-ray absorption near edge structure (XANES) measurements with ~ 30 -50 nm resolution in the 6 – 11 keV photon energy range [19]. The beamline is optimized to deliver maximum photon flux at the sample location. The Transmission X-ray Microscope (TXM) has been designed while emphasizing maximum stability, simplicity, spatial compactness, and a modular approach [20]. Fig. 3 depicts a schematic of TXM with an overview photograph and CAD rendering of the sample area. The majority of the motion components are off-the-shelf Attocube ECS-series horizontal and vertical positioners mounted on extended SmarAct CLL-series rail positioners, enabling a large travel range along the X-ray beam direction. Piezo-based

actuation systems were chosen due to their compact size, minimal heat dissipation, and relatively high (above ~200 Hz) resonance frequencies. To perform high-resolution tomographic measurement, a high-quality rotary stage is required. The core of the sample stack is the PI air-bearing rotary stage equipped with a 56-channel Moog slip ring, thus enabling infinite rotation. The run-out measurements were performed on the rotary stage assembly prior to installation and yielded the sub-100 nm run-out peak-to-peak asynchronous errors. The sample centring stages are mounted on top of the rotary stage with their signal wires passing through the slip ring. In total, 14 channels out of 56 are available for user experiments that require electrical connections. Significant emphasis was given to the data acquisition speed, and due to large photon flux at the sample location (3.9×10^{13} photons/s at condenser entrance), full tomography sets could be acquired in less than 1 minute with ~30-50 nm spatial resolution [19] and about 10 seconds using AI-based methods [21]. It is also noteworthy to mention that due to the compact and modular design of the instrument, the measured sample image drift over a 12-hour time period did not exceed ~100 nm.

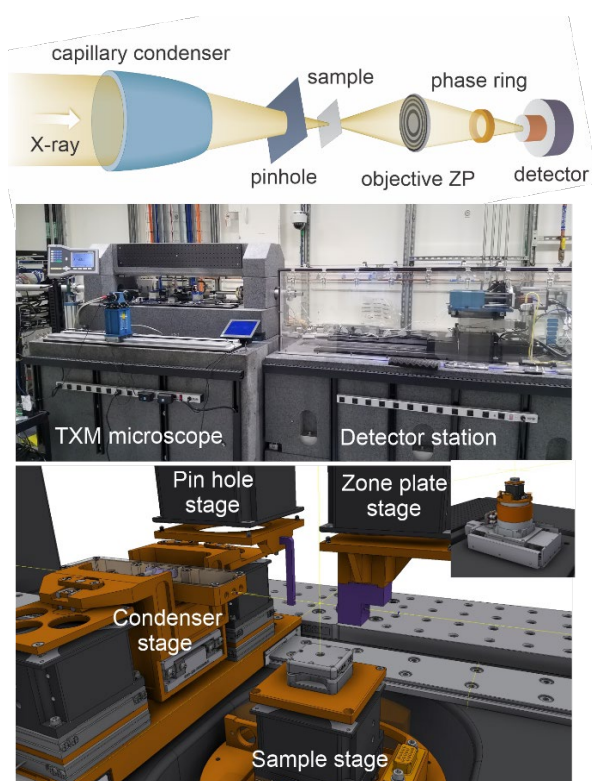


Figure 3: Top panel: schematic of a TXM microscope listing all key components. Middle panel: a photo of the microscope and the detector station installed at 18-ID. Lower panel: CAD model of capillary condenser/pinhole/sample stage/zone plate assembly.

CONCLUSION

In summary, a brief overview that covered microscopy instrumentation developments at NLSL-II during the past decade has been provided. By leveraging the in-house

expertise in microscopy instrumentation and nano-focusing optics development, several cutting-edge, high-performing systems have been designed, constructed, and commissioned. As focusing optics and advanced control systems continue to evolve rapidly, greater synergy between precision engineering, control algorithms, and optics fabrication is anticipated, allowing delivery of adequate microscopy instrumentation for the 4th generation synchrotron facilities.

ACKNOWLEDGEMENTS

This work used resources of the National Synchrotron Light Source II, a U.S. Department of Energy (DOE) Office of Science User Facility operated for the DOE Office of Science by Brookhaven National Laboratory under Contract No. DE-SC0012704.

REFERENCES

- [1] A. Michelson *et al.*, “Three-dimensional visualization of nanoparticle lattices and multimaterial frameworks”, *Science*, vol. 376, no. 6589, pp. 203-207, 2022. doi:10.1126/science.abk0463
- [2] A. Ulvestad *et al.*, “Multimodal X-ray imaging of grain-level properties and performance in a polycrystalline solar cell”, *J. Synchrotron Radiat.*, vol. 26, pp. 1316-1321, 2019. doi:10.1107/s1600577519003606
- [3] I. Calvo-Almazán *et al.*, “Strain Mapping of CdTe Grains in Photovoltaic Devices”, *IEEE J. Photovoltaics*, vol. 9, no. 6, pp. 1790-1799, 2019. doi:10.1109/JPHOTOV.2019.2942487
- [4] F. T. Ling *et al.*, “Nanospectroscopy Captures Nanoscale Compositional Zonation in Barite Solid Solutions” *Sic. Rep.*, vol. 8, no. 13041, 2018. doi:10.1038/s41598-018-31335-3
- [5] E. Nazaretski *et al.*, “Design and performance of an X-ray scanning microscope at the Hard X-ray Nanoprobe beamline of NLS-II”, *J. Synchrotron Radiat.*, vol. 24, pp. 1113-1119, 2017. doi:10.1107/S1600577517011183
- [6] H. Yan *et al.*, “Multimodal hard X-ray imaging with resolution approaching 10 nm for studies in material science”, *Nano Futures*, vol. 2, no. 1, 2018. doi:10.1088/2399-1984/aab25d
- [7] E. Nazaretski *et al.*, “Pushing the limits: an instrument for hard X-ray imaging below 20 nm”, *J. Synchrotron Radiat.*, vol. 22, pp. 336-341, 2015. doi:10.1107/S1600577514025715
- [8] E. Nazaretski *et al.*, “Performance and characterization of the prototype nm-scale spatial resolution scanning multilayer Laue lenses microscope”, *Rev. Sci. Instrum.*, vol. 84, no. 3, p. 033701, 2013. doi:10.1063/1.4774387
- [9] E. Nazaretski *et al.*, “Design and performance of a scanning ptychography microscope”, *Rev. Sci. Instrum.*, vol. 85, p. 33707, 2014. doi:10.1063/1.4868968
- [10] R. Conley *et al.*, “Multilayer Laue Lens: A Brief History and Current Status”, *Synchrotron Radiat. News*, vol. 29, no. 4, pp. 16-20, 2016. doi:10.1080/08940886.2016.1198669
- [11] Saša Bajt *et al.*, “Achieving focusing with efficient high-NA multilayer Laue lenses”, *Light Sci. Appl.*, vol. 7, p. 17163, 2018. doi:10.1038/lsa.2017.162

- [12] N. Bouet *et al.*, “Large Aperture and Wedged Multilayer Laue Lens for X-ray Nanofocusing”, *J. Nanoscience and Nanotechnology*, vol. 19, no. 1, pp. 575-584, 2019. doi:10.1166/jnn.2019.16479
- [13] H. Yan, X. Huang, N. Bouet, J. Zhou, E. Nazaretski and Y. S. Chu, “Achieving diffraction-limited nanometer-scale x-ray point focus with two crossed multilayer Laue lenses: alignment challenges”, *Opt. Express*, vol. 25, no. 21, pp. 25234-25242, 2017. doi:10.1364/OE.25.025234
- [14] Weihe Xu *et al.*, “A versatile high-speed x-ray microscope for sub-10 nm imaging”, *Rev. Sci. Instrum.*, vol. 95, p. 113705, 2024. doi:10.1063/5.0225904
- [15] E. Nazaretski *et al.*, “A new Kirkpatrick–Baez-based scanning microscope for the Submicron Resolution X-ray Spectroscopy (SRX) beamline at NSLS-II”, *J. Synchrotron Radiat.*, vol. 29, p. 1284, 2022. doi:10.1107/S1600577522007056
- [16] X. Liu *et al.*, “Environmentally Friendly Zr-Based Conversion Nanocoatings for Corrosion Inhibition of Metal Surfaces Evaluated by Multimodal X-ray Analysis”, *ACS Appl. Nano Mater.*, vol. 2, no. 4, pp. 1920–1929, 2019. doi:10.1021/acsanm.8b02309
- [17] A. Mohanty *et al.*, “The effect of copper additives on hybrid Zr-based chemical conversion coating morphology and chemical compositions”, *Appl. Surf. Sci. Adv.*, vol. 27, p. 100726, 2025. doi:10.1016/j.apsadv.2025.100726
- [18] J. Kim, K. Lauer, H. Yan, Y. S. Chu and E. Nazaretski, “Compact prototype apparatus for reducing the circle of confusion down to 40 nm for x-ray nanotomography”, *Rev. Sci. Instrum.*, vol. 84, p. 035006, 2013. doi:10.1063/1.4798546
- [19] M. Ge *et al.*, “One-minute nano-tomography using hard X-ray full-field transmission microscope”, *Appl. Phys. Lett.*, vol. 113, p. 083109, 2018. doi:10.1063/1.5048378
- [20] D. S. Coburn *et al.*, “Design, characterization, and performance of a hard x-ray transmission microscope at the National Synchrotron Light Source II 18-ID beamline”, *Rev. Sci. Instrum.*, vol. 90, p. 053701, 2019. doi:10.1063/1.5088124
- [21] J. Zhang W-K. Lee and M. Ge, “Sub-10 second fly-scan nano-tomography using machine learning”, *Communications Materials*, vol. 3, p. 91, 2022. doi:10.1038/s43246-022-00313-8